



Mission: Your First Power App

A Field Guide for Citizen Developers

This guide provides the strategic and tactical knowledge required to successfully plan, build, and deploy your first mission-critical Power App.

THE MISSION: RAPID APPLICATION DEVELOPMENT

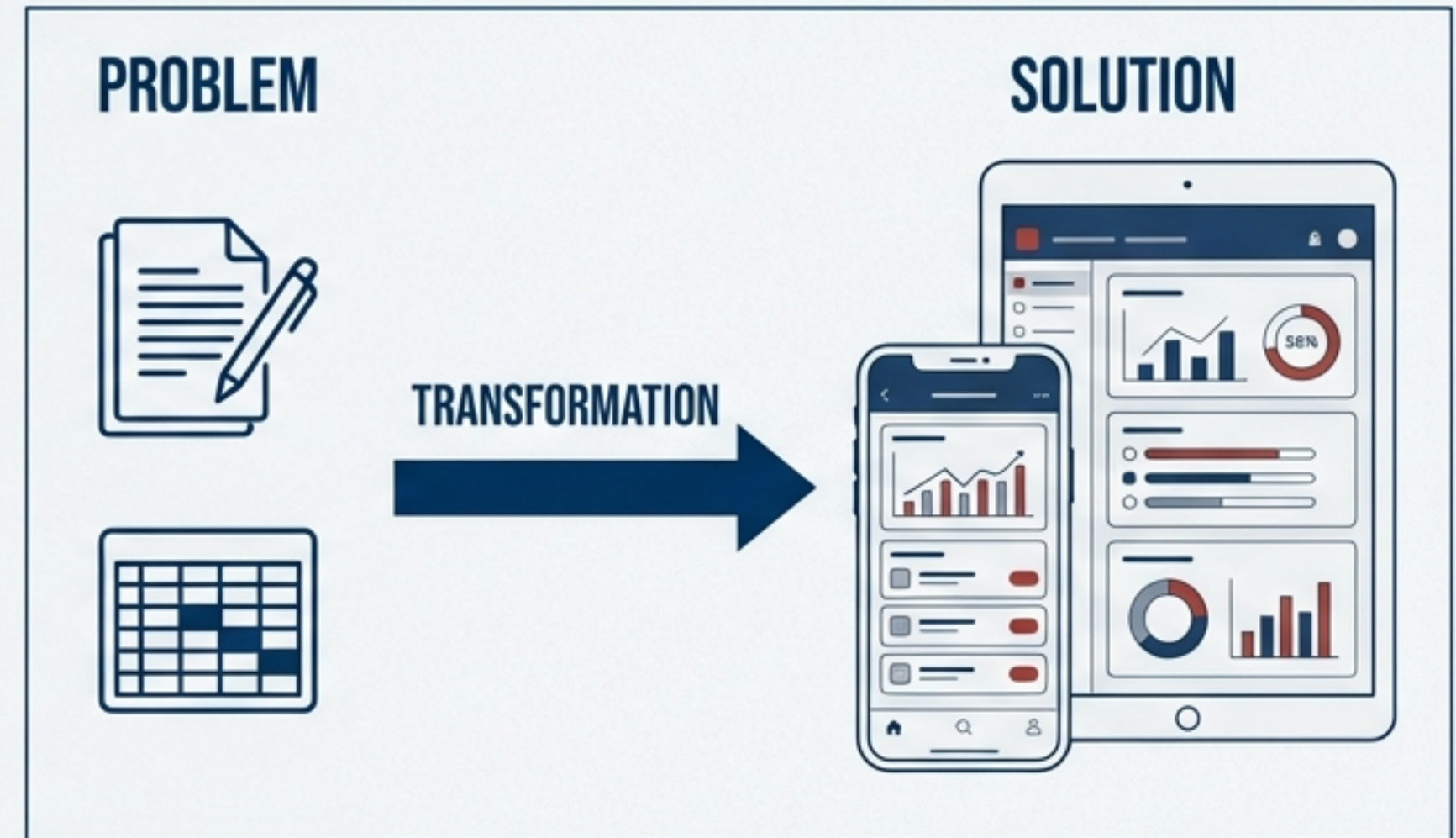
Power Apps is a low-code platform that allows you to build custom applications to solve everyday problems without extensive coding knowledge. Think of it as a force multiplier for your unit's efficiency.

WHAT CAN YOU BUILD?

- * **Trackers:** Equipment checkout, personnel accountability, task completion.
- * **Forms:** Digital versions of paper forms for inspections, requests, and reports.
- * **Information Hubs:** Centralized portals for key documents and contacts.

KEY OBJECTIVE

Empower you to transform manual, inefficient processes into streamlined digital solutions, saving time and reducing errors.



COMMANDER'S INTENT: 5 PRINCIPLES FOR A SUCCESSFUL APP



Principle 1: Make the User the Star

- Building in isolation is a common failure point. You might know the mission, but the end-users know the tactical realities of the process.
- Involve the actual users of the app in the planning, requirements gathering, and testing phases.
- **Result:** The users become your “Champions.” They help with adoption, train their peers, and are emotionally invested in the app's success.

Principle 2: Fix the Process First

- A Power App cannot fix a broken or non-existent business process. You cannot build the airplane while it's in flight.
- First, define a structured, repeatable process. Know exactly how you want it to work from start to finish.
- **Result:** The app amplifies and enforces a good process, rather than creating digital chaos around a bad one.

COMMANDER'S INTENT (CONT'D)

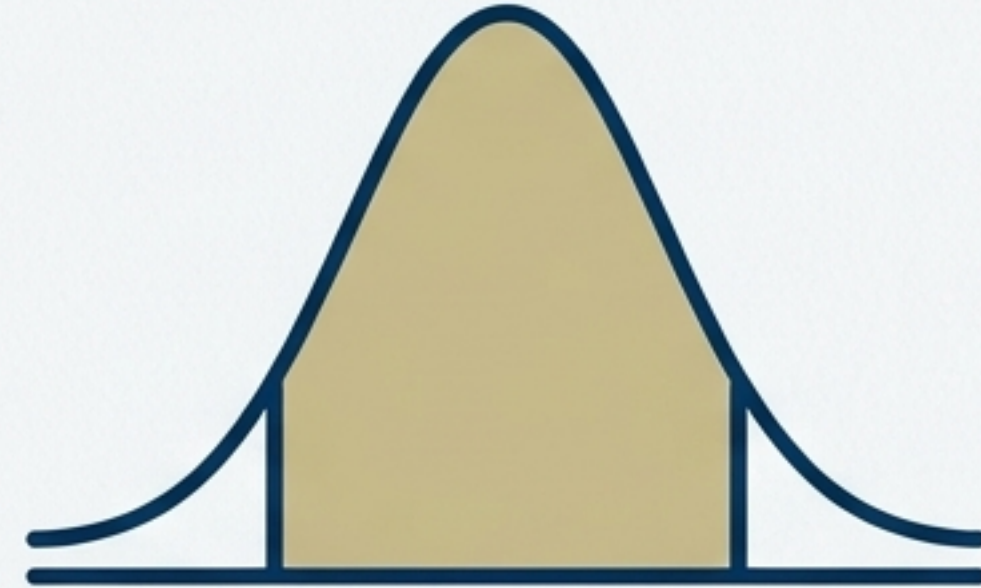


Principle 3: Employ the 'Hulk Smash' Mentality

Avoid the temptation to build one "Mega App" that does everything. Large apps suffer from poor performance, complex navigation, and are a single point of failure.

Instead, build smaller, highly focused apps. An app should do one thing and do it really well.

Example: Build one app for service members to enter time in the field (mobile-focused) and a separate app for project managers to approve time (desktop-focused). Same data, different missions.



Principle 4: Build for the 99 %, Not the Unicorn

Don't design your app around rare, one-off scenarios (the "unicorns"). This adds massive complexity for minimal gain.

Focus on the daily use case. Just because you *can* make an app work offline or be perfectly responsive on every device doesn't mean you *should* if it triples the development time for a feature that's rarely needed.

COMMANDER'S INTENT (CONT'D)



Principle 5: Define the Finish Line

Without a clear definition of “done,” projects suffer from “scope creep.” The finish line keeps moving. Before you start, define what a successful Phase 1 looks like.

Example: “Phase 1 is complete when the app can successfully create, read, update, and delete equipment checkout records.”

Result: You achieve a sense of accomplishment and can roll out value in stages. You can always start a new “race” (Phase 2) after you cross the first finish line.

INTEL GATHERING: CHOOSING YOUR DATA SOURCE

Your app needs a place to store data. For Power Apps, the two most common starting points are SharePoint Lists and Microsoft Dataverse.

Feature	 Microsoft Dataverse	 SharePoint Lists
Data Modeling	Relational, normalized (like a true database)	Flat lists, basic lookups
Security	Robust: Row & field-level security, audit logs	Basic: Item-level permissions only
Licensing	Requires a Premium Power Apps plan	Included in most Microsoft 365 plans
Performance	Highly delegation-friendly, scales to massive datasets	Limited delegation (queries over 2000 records may fail)
Ideal For	Complex, scalable, enterprise-grade apps	Simple, content-based, team-focused apps

Commander's Recommendation for this Mission:
Start with **SharePoint Lists**. It's included in your existing M365 license, is easy to set up, and is perfect for building your first few applications.

MISSION PREP: ESTABLISH YOUR DATA SOURCE IN SHAREPOINT

Before building the app, we need to create our data source. We will build an app to track equipment checkouts.

Action Steps:

1. Navigate to your unit's SharePoint site.

2. Click **New > List**.

3. Select **Blank list**. Name it `Equipment Checkout` and click **Create**.

4. Add the following columns by clicking **+ Add column**:

- **Title** (Rename this to `Equipment Name` - Single line of text, Required)

- **`Checked Out To`** (Person or Group)

- **`Checkout Date`** (Date and time)

- **`Status`** (Choice column with options: `Available`, `Checked Out`, `Maintenance`)

- **`Notes`** (Multiple lines of text)

Your SharePoint list should look like this:

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Equipment Checkout

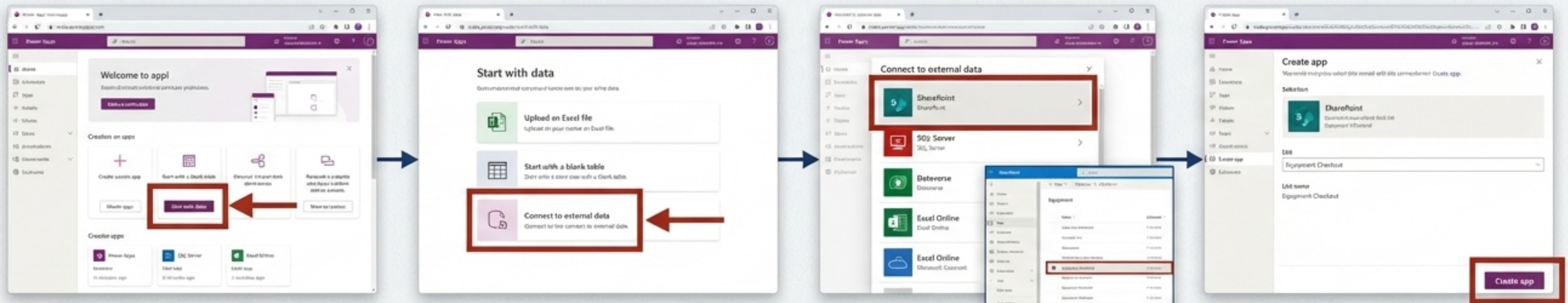
Equipment Checkout

Equipment Name	Checked Out To	Checkout Date	Status	Notes
Toughbock CF-33	Sgt. Miller	10/26/2023 08:00 AM	Checked Out	
Toughbock CF-33	Sgt. Miller	10/26/2023 08:00 AM	Checked Out	Screen protector has a minor scratch.
PRC-1176 Rsdio	A1C Davis	10/25/2023 14:30 PM	Checked Out	
Night Visien Geggiss			Available	
Satsillite Phone			Maintenance	Battery not holding s charge.

EXECUTION PHASE I: GENERATE THE APP FROM DATA

Power Apps can build a fully functional, three-screen app for you in seconds.

Practical Exercise:



1. Select **Start with data**.
2. Choose to connect.
3. Select SharePoint.
4. Click **Create app**.

After about 30 seconds, Power Apps Studio will open with your new, fully functional application.

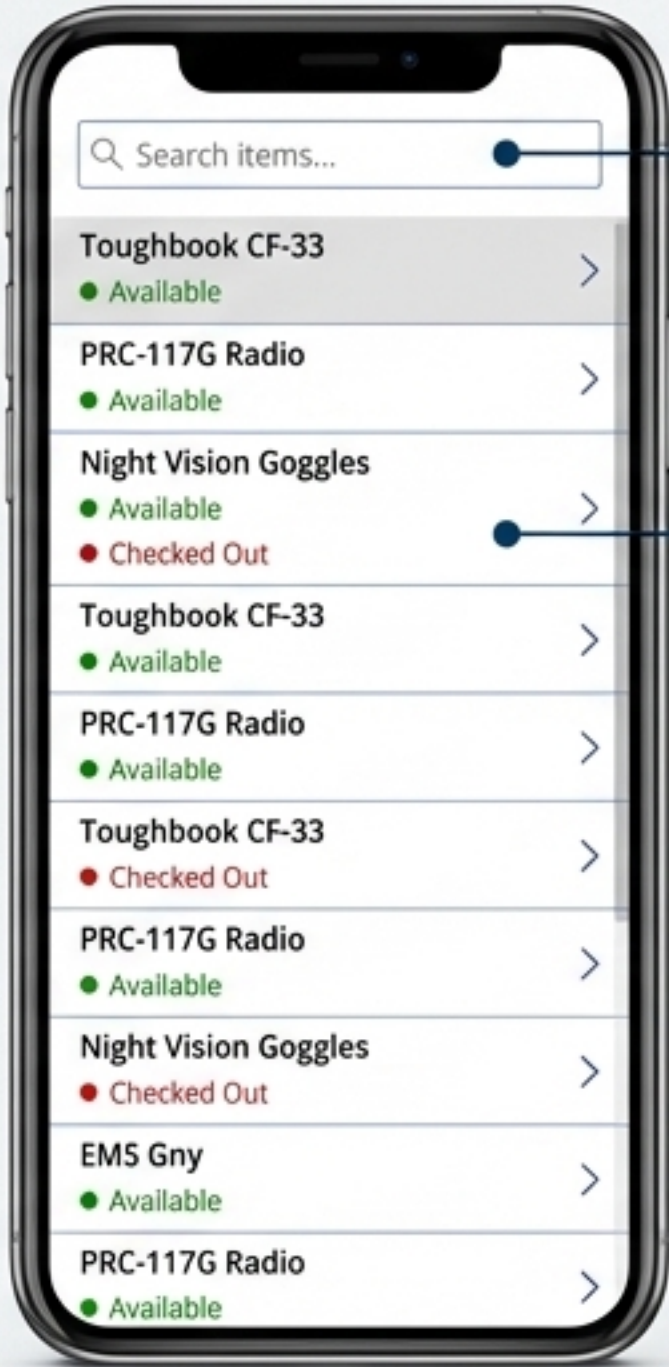
INITIAL RECON: ANATOMY OF YOUR NEW APP

Your auto-generated app has three main components that handle all the basic functions: Create, Read, Update, and Delete (CRUD).

1. The Gallery ('BrowseGallery1')

This control displays a **list of records** from your data source. It's a window into your SharePoint list.

You configure the **top row (template row)**, and all other rows automatically follow the same format.



3. The Search Box ('TextSearchBox1')

A simple text input that allows users to filter the items shown in the gallery.



2. The Form ('DetailForm1' & 'EditForm1')

This control is used for viewing or editing a **single, specific record**.

When you select an item in the gallery, its details are shown in the form.



Test it out! Click the 'Play' button (▶) in the top right to preview the app and see how these components work together.

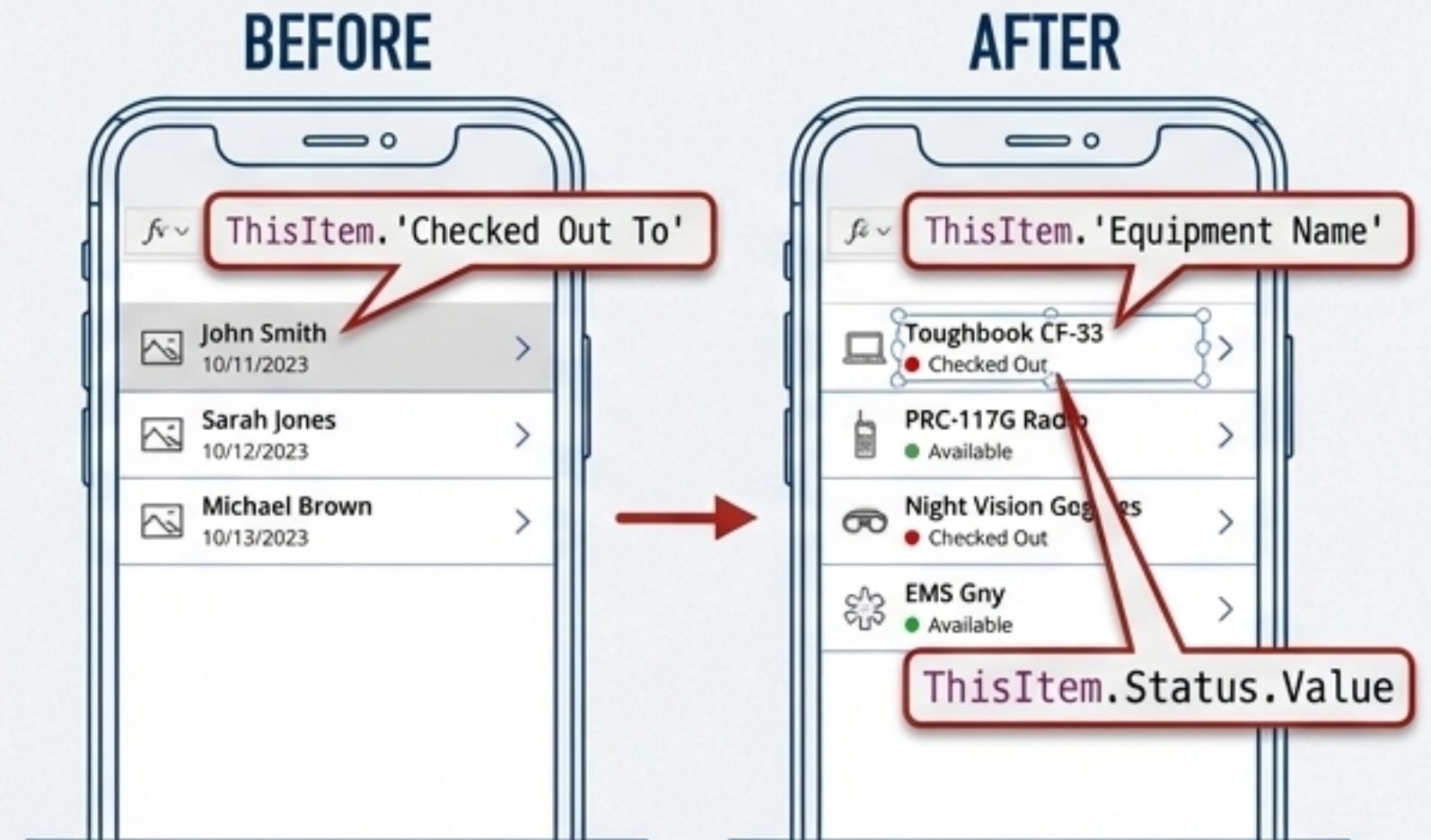
PRACTICAL EXERCISE: CUSTOMIZE THE GALLERY DISPLAY

Objective:

Change the gallery to display the 'Equipment Name' and its current 'Status'.

1. In the Tree View on the left, select **BrowseGallery1**.
2. Click on the **top row** of the gallery to select the template. You may need to click a specific label inside the template row.
3. With the top label selected (it likely shows "ThisItem.'Checked Out To'"), find the **Text** property in the formula bar at the top of the screen.
4. Change the formula to:
`ThisItem.'Equipment Name'`
5. Select the label below it. Change its formula to:
`ThisItem.Status.Value`

(Note: For a "Choice" column, you need to add **.Value** to display the text.)



Power Fx Insight

Power Fx is the formula language of Power Apps. It's very similar to Excel. You reference data using `ControlName.Property` or `ThisItem.ColumnName`.

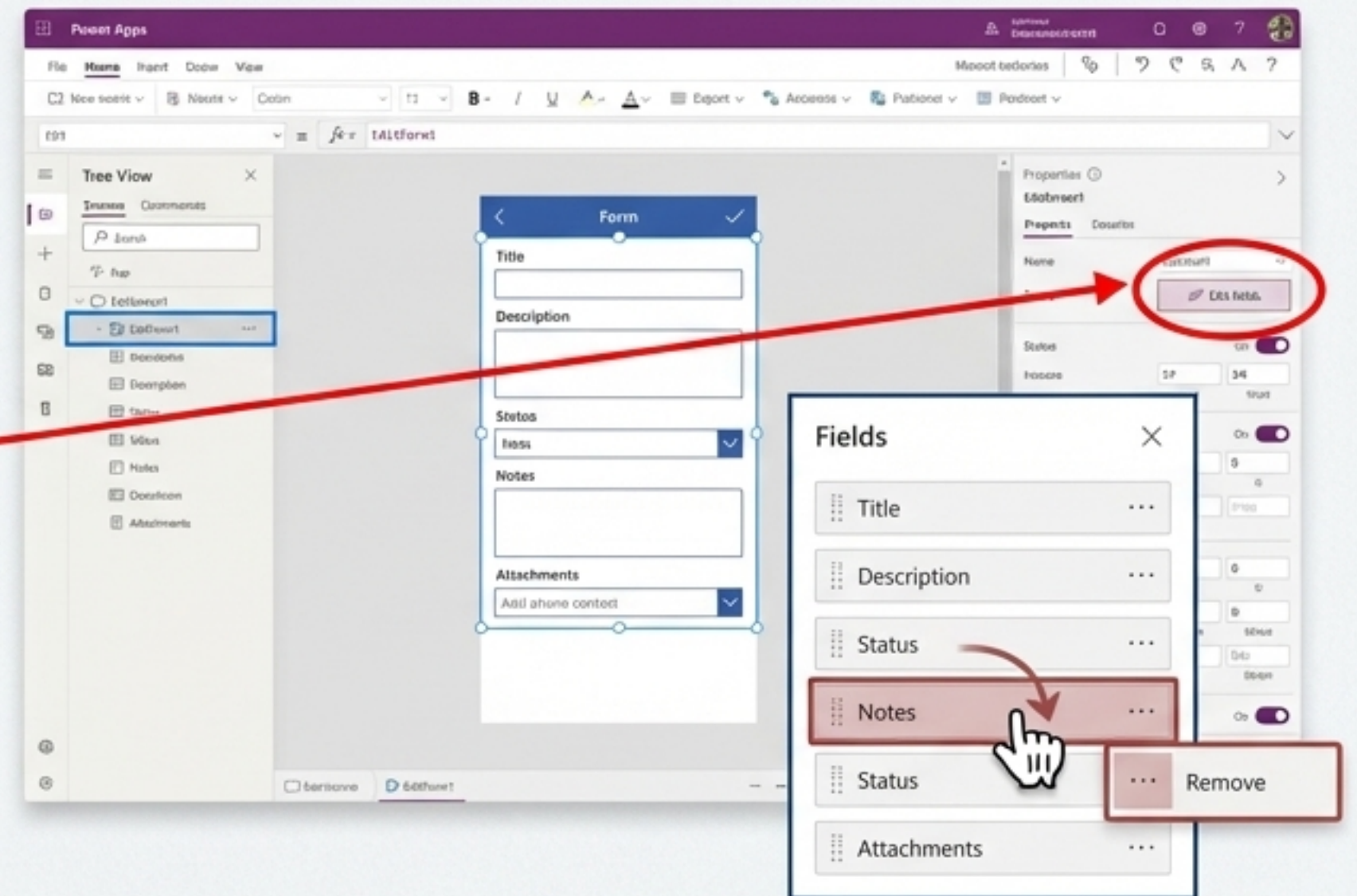
PRACTICAL EXERCISE: MODIFY THE EDIT FORM

Now, let's customize the screen where users will enter and edit data.

Objective:

Move the 'Notes' field to the bottom and remove the 'Attachments' field.

1. In the **Tree View**, expand **EditScreen1** and select the form control, **EditForm1**.
2. In the **Properties** pane on the right, click **Edit fields**.
3. A list of all form fields will appear.
 - To reorder a field, click and drag the **Notes** field to the bottom of the list.
 - To remove a field, click the ellipsis (...) next to **Attachments** and select **Remove**.
4. The form on the canvas will update instantly.



These simple changes allow you to create a cleaner, more intuitive data entry experience for your users.

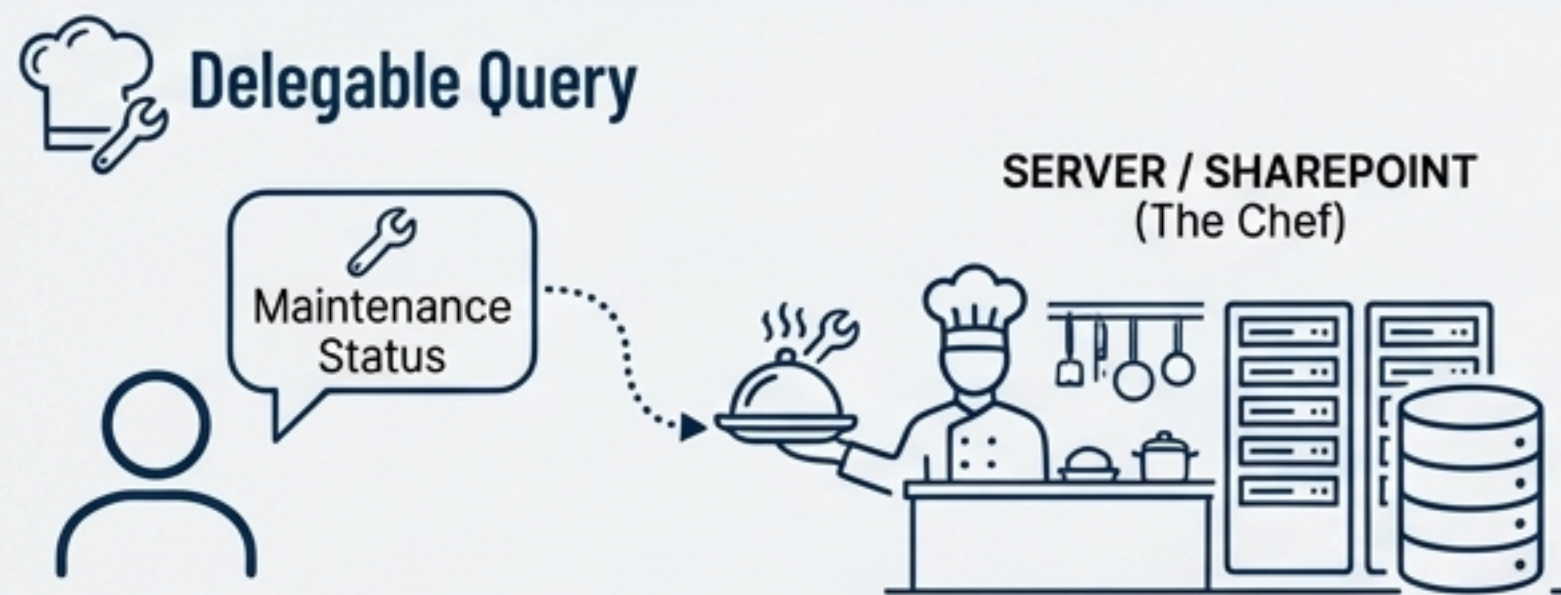
CRITICAL INTEL: UNDERSTANDING DELEGATION

The Problem Statement

How does your app handle large data sources with thousands of records? By default, Power Apps will only process the **first 500 records** locally if a query cannot be "delegated." You can increase this limit to 2,000, but the core issue remains.

What is Delegation?

Delegation is like asking the server (the chef) to do the heavy lifting for you.



"Hey SharePoint, filter your millions of records and only send me the ones for the 'Maintenance' status." (Efficient)



"Hey SharePoint, send me all your records. I'll filter them myself on my phone." (Inefficient, and Power Apps will only bring back the first **500-2000 records**).



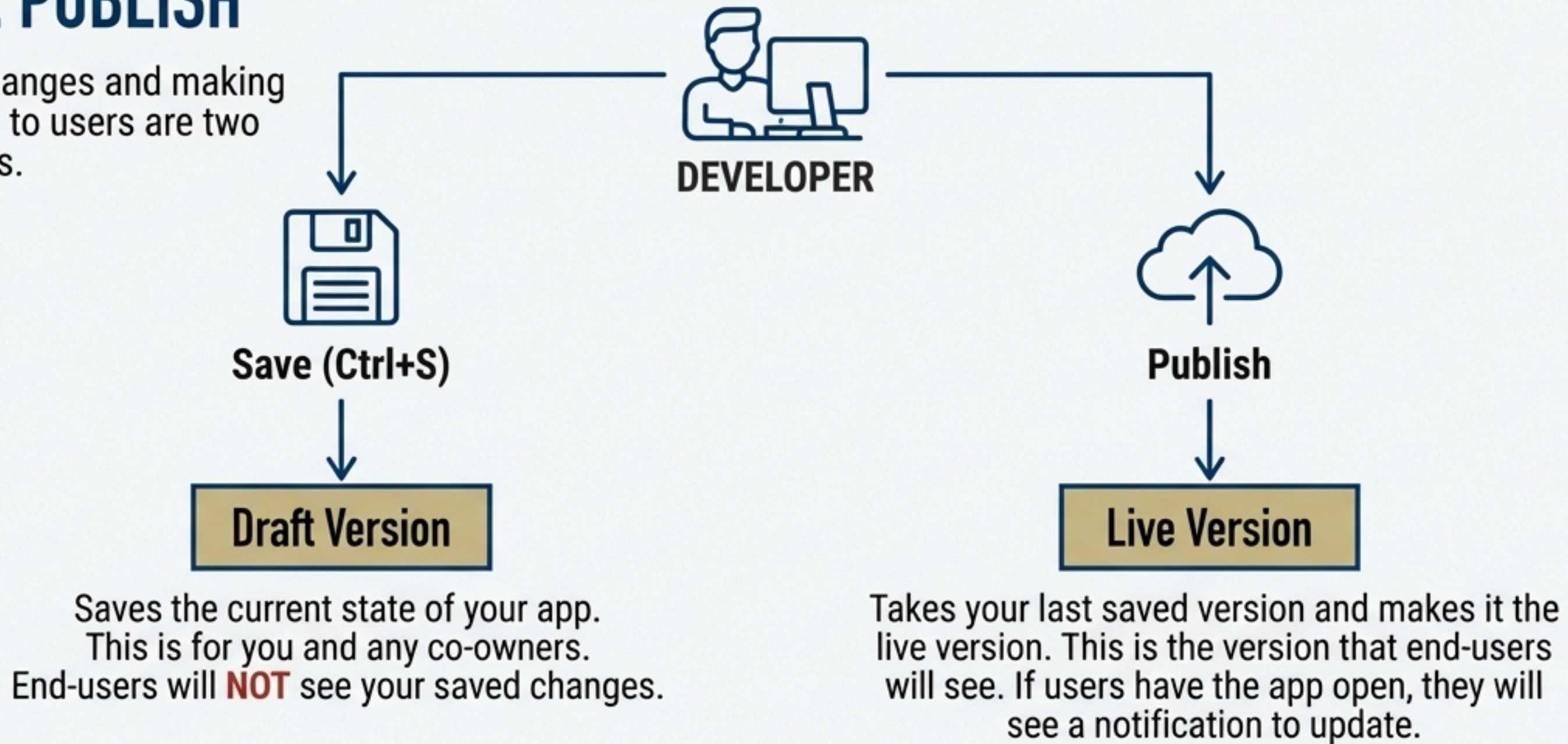
What to Look For Block

Pay attention to the **yellow warning triangle** and **blue underlines** in your formulas. This is Power Apps telling you that a function isn't delegable and might not work correctly on large datasets.

Key takeaway: For now, stick to simple **Filter**, **Search**, and **SortByColumns** functions on simple columns. As you advance, you'll learn more about which functions delegate.

DEPLOYMENT PHASE I: SAVE VS. PUBLISH

Saving your changes and making them available to users are two distinct actions.



Mission Protocol

Always **Save** your progress frequently. When you have tested your changes and are ready for users to see them, click **Publish this version**.

DEPLOYMENT PHASE II: SHARING AND PERMISSIONS

An app is only useful if your team can access it. Sharing involves two critical steps.

Step 1: Share the App

1. From the app details screen or the studio, click **Share**.
2. Enter the names of individuals or type "Everyone" to share with the entire organization.
3. Assign a role:
 - **User**: Can view and use the app.
 - **Co-owner**: Can edit, share, and manage the app.
4. Click **Share**. An email notification will be sent.



Step 2: Share the Data Source (CRITICAL)

Sharing the app **DOES NOT** automatically grant permissions to the underlying data.

You must also navigate to your 'Equipment Checkout' SharePoint list and share it with the same users, giving them at least "Contribute" permissions so they can add and edit records.

If you skip this step, users will open the app but see no data and get permission errors.



MISSION DEBRIEF & YOUR NEXT STEPS

Mission Accomplished: You have successfully planned, built, customized, and deployed a functional Power App.

Key Skills Acquired	Your Next Mission
 Strategic Planning: Applying the 5 principles of successful app development.	The journey doesn't end here. Continue to build your skills.
 Data Preparation: Setting up a SharePoint List as a data source.	 Explore Controls: Experiment with other controls like Date Pickers, Dropdowns, and Buttons.
 App Generation: Creating a functional app from data in minutes.	 Add Automation: Learn the basics of Power Automate to send email notifications or start approval processes from your app.
 Customization: Modifying galleries and forms to meet mission needs.	 Practice: The best way to learn is by building. Find another manual process in your unit and build a small, focused app to solve it.
 Deployment: Saving, publishing, and sharing your app securely.	